

1. f 在 0 处可导 $f(0)=0$ 求 $\lim_{x \rightarrow 0} \frac{f(\sin x)}{x}$

$$\lim_{x \rightarrow 0} \frac{f(\sin x)}{x}$$

$$= \lim_{x \rightarrow 0} \frac{f'(\sin x) \cdot \cos x}{1}$$

$$= f'(0)$$

2. $f(x) = \begin{cases} ax^2 + bx + 1, & x \leq 1 \\ \ln x + c(x-1), & x > 1 \end{cases}$

f 在 $x=1$ 处可导

$$f(1) = a + b + 1$$

$$f_-(1) = \lim_{x \rightarrow 1^-} f(x) = a + b + 1$$

$$f_+(1) = \lim_{x \rightarrow 1^+} f(x) = 0$$

$$\Rightarrow a + b = -1$$

$$f'_-(1) = \lim_{x \rightarrow 1^-} \frac{f(x) - f(1)}{x - 1}$$

$$= \lim_{x \rightarrow 1^-} \frac{ax^2 + bx + 1 - (a + b + 1)}{x - 1}$$

$$= \lim_{x \rightarrow 1^-} (2ax + b)$$

$$= 2a + b$$

$$f'_+(1) = \lim_{x \rightarrow 1^+} \frac{f(x) - f(1)}{x - 1}$$

$$= \lim_{x \rightarrow 1^+} \frac{\ln x + c(x-1) - 0}{x - 1}$$

$$= \lim_{x \rightarrow 1^+} \left(\frac{1}{x} + c \right)$$

$$= c + 1$$

$$\Rightarrow 2a + b = c + 1$$

$$3. f(x) = \begin{cases} \frac{\sin(ax)}{x}, & x \neq 0 \\ b, & x = 0 \end{cases}$$

使 f 在 $x=0$ 处可导

$$f(0) = b$$

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{\sin(ax)}{x}$$

$$= a$$

$$\Rightarrow a = b$$

$$f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{\sin(ax)}{x} - b}{x}$$

$$= \lim_{x \rightarrow 0} \frac{\sin(ax) - bx}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{\cos(ax) \cdot a - b}{2x}$$

$$\Rightarrow x \rightarrow 0 \text{ 时, } \cos(ax) \cdot a - b \rightarrow 0$$

$$\Rightarrow a - b \rightarrow 0, a = b$$

$$4. \lim_{h \rightarrow 0} \frac{f(1+2h) - f(1)}{h} = 6$$

求 $y = f(x)$ 在 $(1, 2)$ 切线方程

$$f(1) = 2$$

$$\lim_{h \rightarrow 0} \frac{f(1+2h) - 2}{h} = 6$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{f(1+2h) - 2}{2h} = 3$$

$$\Rightarrow f'(1) = \lim_{2h \rightarrow 0} \frac{f(1+2h) - f(1)}{2h - 0}$$

$$= 3$$

$$\Rightarrow y - 2 = 3(x - 1)$$